STA301-STATIC

FINAL TERM SOLVED MCQS

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ALL answers are verified if found any mistake then Correct ACCORDINGLY

1. **The marginal p.d.f of any continuous variable of obtained by ………………...**
   a. Averaging
   b. Integrating
   c. Multiplying
   d. Dividing
2. **in a normal distribution how area lies between $\mu \pm \sigma$**
   a. 65%
   b. 68.26%
   c. 75%
   d. 80%
3. **A correlation coeffient**
   a. Tells you the direction of the slope of the scttergram
   b. Is a sort of index how of close the point of a scatter gram deviated from straight line through those points.
   c. Efficiently summaries some of the in scatterplot.
   d. All of these
4. **The values of moment ratios b1 and b2 of normal distribution are:**
   a. 0 and 1
   b. 0 and 2
   c. O and3
   d. 0 and 4
5. **Which of the following is not a property of a binomial experiment?**
   a. The experiment consists of a sequence of an identical trials
   b. Each outcome can be referred to as a success or a failure
   c. The probabilities of the two outcomes can change from one trial to the next.
   d. All of the above
6. **we can obtain the individual probabilities of X and Y from the joint probability function / distribution of (x, y) such individual probabilities are kown as**
   a. Marginal probabilities

b. Bivariate probabilities
c. Separate probabilities
d. Independent probabilities

7. Assume a large distribution in which 6 = 6. What is the standard error of the the mean based for which n = 10?

a. 167
b. 15
c. 3
d. 1.9

8. In a bivariate probability distribution of X and Y,

If f(1, 1)= 4/15, g (1) =2/3 , h(1)=2/5

Then

a. X and Y are statistically independent
b. X and y are not statistically independent
c. Both of these
d. None of these

9. In hyper geometric distribution events can be classified in to

a. A single category
b. 2 categories
c. 3 categories
d. 4 categories

10. In binomial distribution, formula of calculating mean is:

a. $\mu = p + q$
b. $\mu = np$
c. $\mu = p\ q$
d. $\mu = q\ n$

11. **Poisson distribution can be used to approximate the hyper geo metric distribution when , n<0.05, n>20 and**

a. P>0.05
b. P<0.05
c. P>0.10
d. P<0.10

12. the hypergeomertric probability distribution has parameters.

a. N, n
b. N, P, n

c. N, n, k
d. N, N- k, n

13. A random experiment is one is a one in which we repeat it a large number of times under similar condition an it produces.

a. Same result
b. Different result
c. Independent result
d. None of the above

14. Given below is the probability distribution of a random variable X, what is the value of "K"

X	0	1	2	3	4
P(x)	0.5	0.2	0.07	k	0.03

a. 0.02
b. 0.1
c. 0.20
d. 0.002

15. Poisson distribution is never:

a. Symmetrical
b. Positively skewed
c. Negatively skewed
d. Asymmetrical

16. if X and Y are independent, then Cov (X, y) is equal to:

a. 0
b. 1
c. 0 and 1
d. None

17. if E(x) = 3 then E (4X + 3)=……………..

a. 15
b. 12
c. 8
d. 3

18. Which of the following is not a requirement of a binomial distribution?

a. A constant probability of success
b. Only two possible outcomes

c. A fixed number of trials
d. Equally likely outcomes

19. Which of the following is the requirement of a uniform distribution?

a. A constant probability of success
b. Only two possible outcomes
c. A fixed number of trials
d. Equally likely outcomes

20. if we are testing H0 : mean = 250 against
H1: mean > 250 (exceeds 250). Then Rejection region will be

a. In the center
b. On both sides of mean
c. On left side of mean
d. On right side of mean

21. Suppose H0: u 1< u 1 H1: u 1 > u 2 and critical value is value = 1.645 if calculated value of Z = 1.88 then what will be your conclusion?

a. Accept Ho
b. Reject Ho
c. Impossible to decide
d. None of the above

22. The t-distribution can never become narrower than:

a. F-distribution
b. Standard normal distribution
c. Chi-square distribution
d. Exponential distribution

23. t-distribution can be used to find the confidence interval for:

a. Population variance
b. Population mean
c. Difference between population proportions
d. Populating proportion

24. Which one the following provides the basis for hypothesis testing?

a. Null hypothesis
b. Alternative hypothesis
c. Critical value
d. Test – statistic

25. for test hypothesis Ho μ1 = μ2 and H1: μ1 < μ2, the critical region at 0.05 level of significance and n>30

a. Z less and equal 1.96
b. Z > 1.96
c. Z > 1.645
d. Z<-1.645

26. Alpha is the probability of……..

a. Making type I error
b. Making Type II error
c. Accepting Ho when it is true
d. Rejecting H1 when it is wrong

27. The test statistic to test the U1 = U2 (U represent the mean of population normal population when n > 30.

a. Z- test
b. T – test
c. F – test
d. All of above

28. Suppose that the workers of factory B belives that the average income that the average income of the workers of factory A exceed that average income the null hypothesis will be

a. MeanA > MeanB
b. MeanA ⊇MeanB
c. MeanA< MeanB
d. MeanA ⊆MeanB

29. which of the following statement is NOT true?

a. The t distribution is Symmetric about Zero
b. The t distribution is more spread out than the standard normal distribution
c. As the degrees of freedom get smaller , the t-distributions dispersion get smaller
d. The t distribution is mound - shaped

30. A-------- is a range of number inferred from the sample that sample that has a certain probability of including the population parameter over the long run.

a. Hypothesis
b. Lower limit
c. Confidence interval

d. Probability limit

31. The proportion of cigarette smokers in Pakistan is greater than 20% , the null hypothesis Hois,

a. P < 0.20
b. P ≠ 0.20
c. P ⊆ 0.20
d. P> 0.20

32. A null hypothesis is generally denotes by:

a. H0
b. H1
c. H2
d. H3

33. A deserving player was not selected in the cricket team .it is an example of

a. Type-I error
b. Type – II error
c. Type – III error
d. Type – Iv error

34. As sample size goes up, what tends to happen to 95 %confidence interval?

a. They become wider
b. They become more precise
c. They become more precise and narrow
d. They become more narrow

35. to determine a simple size which test lest we use------

a. Z
b. I
c. F
d. None these

36. How can we interpret 90% confidence interval for the mean of the normal population?

a. There are 10 % chances of falling true value of the parameter
b. There are 90%cahnces of falling true value of the parameter
c. There are 100 chances of falling true value of the parameter
d. All are correct

37. **for the smaller values of v, the t distribution will be ------than the standard normal distribution.**
   a. Less spread out
   b. More spread out
   c. Same length
   d. None of these
38. **if we are testing HO : mean = 50 against**

H1: mean < 50 then rejection region will be

   a. In the center
   b. On left side of mean
   c. On right side of mean
   d. On both side
39. **if we are testing HO : p1 < P2 (0r) p1 – p2 <0 against H1 : p1> P2(or) p1- p2 > 0 and the tabulated of Z= 1.645 and Z Calculated value = 2.63 then what will be conclusion\end {gathered}\]**
   a. Reject Ho
   b. Accept HO
   c. None of the above
   d. Impossible to decide
40. **In Binomial distribution the sample size is considered to be sufficiently large, if both np and nq are greater than or equal to**
   a. 3
   b. 5
   c. 10
   d. 15
41. **To the test hypothesis about the difference of means for large simple, what test statistics will be used?**
   a. Chi- square
   b. F
   c. Z
   d. T
42. **When we start hypothesis testing, we always assume that:**
   a. HO is true
   b. HO is false
   c. H1 is true
   d. All are correct

43. to test hypothesis about the different of means for small samples, test statistics will be used:

a. F- test
b. Z- test
c. T-test
d. All of these

44. ideally, the width of confidence interval should be:

a. 0
b. 1
c. 98
d. 100

45. Conventionally, the probability of making a type- I error is denoted by which of following Symbol?

a. ?(theta)
b. A(Alpha)
c. B(beta)
d. S(sigma)

46. in a random sample of 500 men from Lahore city, 300 are found to smoker ; the proportion of smoker is equal to:

a. 0.3
b. 0.4
c. 0.6
d. 0.5

47. if we are testing HO : P1 – p2= 120 against H1 : P1 – P2 > 120 (exceeds 120) then rejection region:

a. In Center
b. On right side Z= 0
c. On left side Z=0
d. Both side Z= 0

48. In testing HO: μ = 100 against HI: μ ≤ 100 at the 10% level of significance, HO is rejected if:

a. The p-value is greater than 0.10
b. The p-value is less than 0.10
c. 100 is contained in the 90% confidence interval
d. The value of the test is static in the acceptance region

49. In the method of moments, how many equations are required for finding two unknown

a. 2
b. 4
c. 1
d. 3

50. To determine a sample size in estimating population mean, we use the _________ value.

a. X
b. Z
c. F
d. T

51. The method of maximum likelihood (ML) is used to find out:

a. Random numbers
b. Sample values
c. Point estimates
d. Correlation coefficient

52. It is the probability of:

a. Accept HO_I HO is true
b. Reject HO_I HO is true
c. Accept HO_I HO is false
d. Reject HO_I HO is true

53. In interval estimation we obtained a ……. of values as an estimate of parameter.

a. Both a and b
b. Range
c. Group
d. None of these

54. We can apply method of Maximum likelihood on:

a. Qualitative variables only
b. Discrete variables only
c. Discrete as well as continuous variables
d. Continuous variables only

55. Inferential statistics involves:

a. Testing of hypothesis

b. All of these
c. Confidence interval
d. Estimation

56. **Sample proportion is a/an _____________ estimator of population mean.**
a. Consistent
b. Unbiased
c. Unbiased and consistent estimator
d. None of these

57. If var (TI) > Var (T2), where TI and T2 are two unbiased estimators, then:
a. TI is more consistent
b. T2 is more consistent
c. TI is more efficient
d. T2 is more efficient

58. The confidence interval become wide and less precise by
a. None of these
b. Increasing the sample size
c. Decreasing the level of significance
d. Decreasing the sample size (by increasing the sample size the confidence interval become more precise)

59. From which of the following methods we can obtain point estimate of the population parameters:
a. Al of the above
b. Maximum likelihood method
c. Method of moments
d. Method of least squares

60. ___________ are equivalent.
a. Type-II error and level of confidence
b. Type-I error and level of significance
c. Type-II error and level of significance
d. Type-I error and level of confidence

61. In a geometric distribution, maximum likelihood estimator (MLE) for proportion (P) is equal to:
a. Sample variance

b. Sample mean
c. Reciprocal of the sample variance
d. Reciprocal of the mean

62. Internal estimation and Confidence interval are:
a. Different
b. Independent
c. Dependent
d. Same

63. “A failed student was passed by the examiner”, is an example of:
a. Type-I error
b. Type-IV error
c. Type-III error
d. Type-II error

64. A confidence interval has a specified probability ___________ of containing the true value of parameter.
a. a
b. a/2
c. 1+a
d. 1-a

65. When sample size is to be considered large, population standard deviation can be replaced by
a. Sample variance
b. Sample Mean
c. Population variance
d. Sample Standard Deviation

66. An industrial designer wants to determine the average amount of time it takes an adult to assemble an “easy to assemble” toy. A sample of 16 times yielded an average time of 19.92 minutes, with a population standard deviation equal to 5.73 assuming normality of assembly times. What is a 95%.
a. (14.1123,22.7277)
b. (17.1123,24.7277)
c. (17.1123,22.7277)
d. (17.1123,30.7277)

67. Suppose that the worker of factory b believes that the average income of the worker of factory A exceed their average income, the hypothesis will be:

a. Mean A > Mean B
b. Mean A ≥ Mean B
c. Mean A < Mean B
d. Mean A ≤Mean B

68. β is the probability of:

a. Reject HO/HO is true
b. Reject HO/HO is false
c. Accept HO/HO is false
d. Accept HO/HO is true

69. An estimator is said to be efficient if it has:

a. Smallest variance
b. Unbiased variance
c. Both (a & b)
d. None of these

70. Conventionally, the probability of making a type-1 error is denoted by which of the following symbol?

a. S(sigma)
b. β(beta)
c. ? (theta)
d. a(alpha)

71. The end points that bound the confidence interval are called:

a. Lower limit
b. Bounded limit
c. Lower and upper limits
d. Upper limits

72. An estimator is said to be _____if its expected value is equal to true value of its parameter.

a. Efficient
b. Consistent
c. Biased
d. un Biased

73. The width of the interval is called ____of the estimate.

a. Biased
b. Accuracy
c. Precision
d. Unbiasedness

74. If we draw a sample of size "n" from a population then sample is called "large sample"it.

a. n>40
b. n>30
c. n>50
d. n>25

75. The precision of an estimates can be increased by increasing the:

a. sample value
b. size of population
c. number of parameters
d. size of sample

76. Which of the following is a most important and widely used method in point estimates?

a. The method of least squares
b. The method of moments
c. The method of maximum likelihood
d. sample value

77. An experiment was conducted to estimate the mean yield of a new variety of wheat. A sample of 20 plots gave a mean yield of 2.9 and a 95% confidence of (2.48, 3.32).

a. None of these
b. We are sure the mean yield of this new Varity is between 2.48and 3.32
c. We are 95% confidence that the true mean yield of this is 2.6
d. We are 95% confident the true mean valid of this variety is between 2.48 and 3.32

78. A ………………….. Is a range of number is inferred from the sample that has a certain probility of including the population parameter over the long run.

a. Hypotheses
b. Lower limit
c. Confidence interval
d. Probability limit

79. To determine a sample size which test we use……

a. t
b. F
c. None of these
d. Z

80. A statistic whose standard error decreases with an increase in the sample size will be…

a. consistent
b. unbiased
c. efficient
d. none

81. The precision of an estimator can be increased by decreasing the:

a. level of significance
b. number of estimators
c. number of parameters
d. size of sample

82. Conventionally, he probability of making a type-error is denoted by:

a. beta
b. theta
c. alpha
d. sigma

83. Sample mean is a ……..estimate of population mean.

a. Biased
b. None of these
c. Unbiased
d. Either unbiased or based

84. If we draw a sample of size 'n' from a population then sample is called 'large sample' if;

a. N>30
b. N>501
c. N>40
d. N>25

85. When sample size is to be considered large population standard deviation can be replaced by

a. Population variance

b. Sample variance
c. Sample mean
d. Sample standard derivation

86. Suppose that the worker of factory B believe that the average income of the worker of factory A exceeds their average income the hypothesis will be
a. MeanA>MeanB
b. MeanA>/MeanB
c. MeanA<MeanB
d. MeanA</MeanB

87. For particular hypothesis test alpha = 0.09 and beta = 0.03 what is the value of type of error
a. 0.09
b. 0.91
c. 0.03
d. 0.97

88. Suppose H0:u1<u2 H1:u1.u2 and critical value is = 1.645if calculated value of z= 1.88then what will be your
a. Accept Ho
b. Reject HO
c. Impossible to decide
d. None of the above

89. Suppose H0: p=0 H1:p,0 and critical value is Z=2.33if calculate the value of Z=-2.60 then what will be your calculation
a. Accept Ho
b. Reject HO
c. Impossible to decide
d. None of the above

90. If the null hypothesis is that the average height of Pakistan soldiers exceed the average height of American soldiers by more than 3 inches what will it's alternative hypothesis
a. p>3
b. p<3
c. P=3
d. None of the above

91. Pairing is done only in case of ….. observation
a. Dependent

b. Independent
c. Small
d. Large

92. The critical region is computed from the value of

a. alpha
b. Beta
c. 1-Alpha
d. 2-Beta

93. When examining group difference where not direction of the difference is specified which of the following is used

a. one tail test
b. Two tail test
c. Directional hypothesis
d. Difference of mode test

94. If we traveling at H0 :mean = 50 against

H1: mean<50 then the rejection region will be

a. In the center
b. On left side of mean
c. On right side of Mean
d. On both sides

95. If our significance level of 5% is used rather than 1% the null hypothesis is

a. Less likely to be rejected
b. Just as likely to be rejected
c. More likely to be rejected
d. None of these

96. The term 1-beta is called

a. Level of the test
b. Power of the test
c. Size of the test
d. Critical region

97. The T test for independent sample assume which of the following

a. Score are independent
b. Scorer dependent
c. Groups have equal medians
d. Alpha is the probability of

98. The level of significance is also called

a. Power of test
b. Type II error
c. Type I letter
d. Acceptance region

99. For a sample p1=0.2,n1=100 and for second sample p2 =0.5 and n2=200the pooled estimator pc is

a. 0.30
b. 0.45
c. 0.40
d. 0.35

100. The location of the critical region depends upon

a. Null hypothesis
b. Alternative hypothesis
c. Value of alpha

101. Meaning of T distribution will not exist when D.F will equal to

a. 1
b. 2
c. 3
d. 4

102. Which of the following is not a criterion for a good problem situation?

a. It must be authentic
b. It must be clearly structured and defined
c. It must be dearly structured and defined
d. It must be appropriate to student of development

103. To test a hypothesis about the difference of mean for small samples with test statistic sticks will be used

a. z- test
b. t-test
c. F-test
d. All of these

104. For a particular data the value of Pearson's coefficient of skewness is greater than zero. What will be the shape of distribution?

a. Negatively skewed
b. J-shaped
c. Symmetrical
d. Positively skewed (Page 109)

105. In measures of relative dispersion unit of measurement is:

a. Changed
b. Vanish
c. Does not changed
d. Dependent

106. The F-distribution always ranges from:

a. 0 to 1
b. 0 to -∞
c. -∞ to +∞
d. **0 to +∞ (Page 312)**

107. In chi-square test of independence the degrees of freedom are:

a. n - p
b. n - p-1
c. n - p- 2
d. n – 2

108. The Chi- Square distribution is continuous distribution ranging from:

a. $-\infty \leq \chi2 \leq \infty$
b. $-\infty \leq \chi2 \leq 1$
c. $-\infty \leq \chi2 \leq 0$
d. **$0 \leq \chi2 \leq \infty$ (Page 307)**

109. If ŷ is the predicted value for a given x-value and b is the y-intercept then the equation of a regression line for an independent variable x and a dependent variable y is:

a. ŷ = mx + b, where m = slope (Page 121)
b. x = ŷ + mb, where m = slope
c. ŷ = x/m + b, where m = slope
d. ŷ = x + mb, where m = slope

110. If X and Y are random variables, then $E(X-Y)$ is equal to:

a. $E(X)+E(Y)$
b. $E(X)-E(Y)$
c. $X-E(Y)$
d. $E(X)-Y$

111. The parameters of the binomial distribution b(x; n, p) are:

a. x & n
b. x & p
c. **n & p** **Page 212**
d. x, n & p

112. **one** If P (E) is the probability that an event will occur, which of the following must be false:

a. P(E)= - 1
b. P(E)=1
c. P(E)=1/2
d. P(E)=1/3

113. The sample variance $S^2 \frac{\sum\left(x-\bar{x}\right)^2}{n}$ is:

a. Unbiased estimator of 2 σ
b. Biased estimator of 2 σ **Page 260**
c. Unbiased estimator of μ
d. None of these

114. If f (x, y) is bivariate probability density function of continuous r.v.'s X and Y then is g (x):

a. $\int_{-\infty}^{\infty} f(x,y)dx$

b. $\int_{-\infty}^{\infty} f(x,y)dy$

c. $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)dxdy$ page199

d. $\int_{a}^{b}\int_{c}^{d} f(x,y)dxdy$

115. The analysis of variance technique is a method for :

a. Comparing F distributions
b. **Comparing three or more means** **Page 320**
c. Measuring sampling error
d. Comparing variances

116. The continuity correction factor is used when:

a. The sample size is at least 5
b. Both *nP* and *n (1-P)* are at least 30
c. A continuous distribution is used to approximate a discrete distribution
d. The standard normal distribution is applied

117. Stem and leaf is more informative when data is :
a. Equal to 100
b. Greater Than 100
c. Less than 100
d. In all situations

118. The branch of Statistics that is concerned with the procedures and methodology for obtaining valid conclusions is called:
a. Descriptive Statistics
b. Advance Statistics
c. **Inferential Statistics Page 237**
d. Sampled Statistics

119. Which of the following is a systematic arrangement of data into rows and columns?
a. Classification
b. **Tabulation**
c. Bar chart
d. Component bar chart

120. In normal distribution
a. 1
b. 2
c. **3 Page 119**
d. 0

121. If you connect the mid-points of rectangles in a histogram by a series of lines that also touches the x-axis from both ends, what will you get?
a. Ogive
b. Frequency polygon
c. **Frequency curve Page 38**
d. Historigram

122. Which one of the following statements is true regarding a population?
a. **It must be a large number of values Page 12**
b. It must refer to people
c. It is a collection of individuals, objects, or measurements
d. It is small part of whole

123. When $Q_1 = 2 \; and \; Q_3 = 4$,what is the value of Median, if the distribution is symmetrical:

a. 1
b. **2**
c. 3
d. 4

124. In a simple linear regression model, if it is assumed that the intercept parameter is equal to zero, then:

a. The regression line will pass through the origin
b. **The slope of the line will also be equal to 0.**
c. The regression line will pass through the point (0,10).
d. The regression line will pass through the point (0,-10).

125. A failing student is passed by an examiner is an example of:

a. Type I error
b. **Type II error**
c. Correct decision
d. No information regarding student exams

126. How to find $p\left(X+Y \le 1\right)$?

a. f(0, 0) + f(0, 1) + f(1, 2)
b. f(2, 0) + f(0, 1) + f(1, 0)
c. f(0, 0) + f(1, 1) + f(1, 0)
d. f(0, 0) + f(0, 1) + f(1, 0)

127. Which dispersion is calculated from all the observations?

a. **Standard deviation**
b. Quartile deviation
c. Rang
d. Coefficient of Rang

128. Standard deviation of the data 7, 7, 7, 7, 7, 7, 7 is

a. 49
b. Sqrt (7)
c. **0**
d. 7

129. Which one is the poor measure of dispersion in open-end distribution?

a. Range
b. Standard deviation
c. Mean deviation
d. Variance

130. Men tend to marry women who are slightly younger than themselves. Suppose that every man married a woman who was exactly 5 years younger than themselves. Which of the following is correct:

a. The correlation is −5
b. The correlation is 5
c. The correlation is 1
d. The correlation is 0

131. Sum of absolute deviations of the values is least when deviations are taken from:

a. Mean
b. Median
c. Mode
d. Geometric mean

132. Which of the following measures of central location is affected most by extreme values:

a. Geometric Mean
b. Median
c. Mean
d. Mode

133. Which of the following is a critical value of Z when $1-a=95\%$ one tailed test:

a. 2.58
b. 1.645
c. 1.25
d. **1.96**

134. The difference between expected value of statistic and parameter is called:

a. Non-sampling error
b. Sampling error
c. Standard error
d. Bias

135. Which one is the formula of range:

a. $x_m - x_0$
b. $x_0 - x_m$
c. $\frac{x_0 - x_m}{2}$

d. $\frac{x_0 + x_m}{2}$ Page80

136. Which one is the formula of Geometric mean for group data:

a. $anti\log\left[\frac{\sum f\log X}{n}\right]$

b. $anti\log\left[\frac{\sum \log X}{n}\right]$

c. $anti\log\left[\frac{\sum \log fX}{n}\right]$

d. $anti\log\left[\frac{\sum \log fX}{\sum_{n-1}^{n} f}\right]$

137. The F-distribution has …… parameter.

a. One
b. No
c. **Two** **Page 312**
d. Three

138. The sample mean $\overline{X} = \frac{\sum X}{n}$ is an unbiased estimator of population mean (μ) because:

a. $E(\overline{X}) = 0$

b. $E(\overline{X}) = \mu$ page259

c. $E(\overline{X}) > \mu$

d. $E(\overline{X}) < \mu$

139. The degrees of freedom for a t-test with sample size 6 is:

a. 1
b. 3
c. **5** **Page 341**

d. 7

140. The difference between statistic and parameter is called:

a. Bias
b. Standard error
c. Sampling error
d. None of above

141. Mean deviation is always:

a. Less than S.D
b. Greater than S.D
c. Greater or equal to S.D
d. Less or equal to S.D

142. Evaluate: (9-4)!

a. 362880
b. **120**
c. 24
d. 6

143. Which formula represents the probability of the complement of event A:

a. $1 + P(A)$
b. **$1 - P(A)$ Page 156**
c. $P(A)$
d. $P(A) - 1$

144. If the sampling distribution of $\bar{X}$ is normal, the interval $\mu_{\bar{x}} \pm 3\sigma_{\bar{x}}$ includes:

a. 99% of the sample means
b. **99.73% of the sample means Page 228**
c. 98% of the sample means
d. 95% of the sample means

145. The probability distribution of a statistic is called the:

a. Population distribution
b. Frequency distribution
c. Sampling distribution
d. Sample distribution

146. A discrete probability function f(x) is always:

a. Non-negative
b. Negative
c. **One Page 168**
d. Zero

147. An expected value of a random variable is equal to:
a. Variance
b. Mean Page 191
c. Standard deviation
d. Covariance

148. The $f(x|1) =$ ____________:
a. $f(1,1)$
b. $f(x,1)$
c. $\dfrac{f(x,1)}{h(1)}$ page198
d. $\dfrac{f(x,1)}{h(x)}$

149. The area under a normal curve between 0 and -1.75 is
a. .0401
b. .5500
c. .4599 Page 230
d. .9599

150. Which of the following is impossible in sampling:
a. Destructive tests
b. Heterogeneous
c. To make voters list
d. None of these

151. Which one of the following statements is true regarding a sample?
a. It is a part of population Page 13
b. It must contain at least five observations
c. It refers to descriptive statistics
d. It produces True value

152. The data for an ogive is found in which distribution?
a. A relative frequency distribution
b. A frequency distribution
c. A joint frequency distribution
d. A cumulative frequency distribution Page 43

153. 10! =…………
  a. 362880
  b. **3628800**
  c. 362280
  d. 362800

154. The curve of the F- distribution depends upon:
  a. Degrees of freedom Page 312
  b. Sample size
  c. Mean
  d. Variance

155. In testing hypothesis, we always begin it with assuming that:
  a. Null hypothesis is true Page 277
  b. Alternative hypothesis is true
  c. Sample size is large
  d. Population is normal

156. When two coins are tossed simultaneously, P (one head) is:
  a. $\frac{1}{4}$
  b. $\frac{1}{2}$
  c. $\frac{3}{4}$
  d. 1

157. When a fair die is rolled, the sample space consists of:
  a. 2 outcomes
  b. 6 outcomes
  c. **36 outcomes Page 145**
  d. 16 outcomes

158. When testing for independence in a contingency table with 3 rows and 4 columns, there are _________ degrees of freedom.
  a. 5
  b. **6 Page 341**
  c. 7
  d. 12

159. The F- test statistic in one-way ANOVA is:
  a. SSW / SSE
  b. MSW / MSE
  c. SSE / SSW
  d. **MSE / MSW**

160. A uniform distribution is defined by:

a. Its largest and smallest value
b. Smallest value
c. Largest value
d. Mid value

161. Which graph is made by plotting the mid-point and frequencies?

a. Frequency polygon **Page 34**
b. Ogive
c. Histogram
d. Frequency curve

162. In a set of 20 values all the values are 10, what is the value of median?

a. 2
b. 5
c. 10
d. 20

163. The variance of the t-distribution is given by the formula:

a. $\sigma^2 = \sqrt{\frac{v}{v-2}}$

b. $\sigma^2 = \frac{v^2}{v-2}$

c. $\sigma^2 = \frac{v^2}{v-1}$

d. $\sigma^2 = \frac{v}{v-2}$ page293

164. Which one is the correct formula for finding desired sample size?

a. $n = \left(\frac{Z_{\alpha/2}\sigma}{e}\right)^2$

b. $n = \left(\frac{Z_{\alpha/2}\sqrt{\sigma}}{e}\right)^2$

c. $n = \left(\frac{Z_{\alpha/2}\bar{X}}{e}\right)^2$

d. $n = \frac{Z_{\alpha/2}\sigma}{e}$

165. E(4X + 5) =__________

a. 12 E (X)

b. **4 E (X) + 5**
c. 16 E (X) + 5
d. 16 E (X)

166. How P(X + Y < 1) can be find:
a. f(0, 0) + f(0, 1) + f(1, 2)
b. f(2, 0) + f(0, 1) + f(1, 0)
c. f(0, 0) + f(1, 1) + f(1, 0)
d. f(0, 0) + f(0, 1) + f(1, 0)

167. In normal distribution M.D. =
a. 0.5 σ
b. 0.75 σ
c. 0.7979 σ
d. 0.6445 σ

168. In an ANOVA test there are 5 observations in each of three treatments. The degrees of freedom in the numerator and denominator respectively are.......
a. 2, 4
b. 3, 15
c. 3, 12
d. **2, 12**

169. A set that contains all possible outcomes of a system is known as
a. Finite Set
b. Infinite Set
c. **Universal Set Page 134**
d. **No of these**

170. A population that can be defined as the aggregate of all the conceivable ways in which a specified event can happen is known as:
a. Infinite population
b. Finite population
c. Concrete population
d. **Hypothetical population Page 12**

171. The number of telephone calls that pass through a switchboard has a Poison distribution with mean equal to 2 per minute. The probability that no telephone calls pass through the switchboard in two consecutive minutes is:
a. 0.2707
b. 0.0517
c. **0.0183**
d. 0.0366

172. The range of the binomial distribution is:

a. 0, 1, 2, ... , 100
b. **0, 1, 2, … , n**
c. 0, 1, 2, … , x
d. 1, 2, … , n

173. Which of the following is NOT CORRECT about a standard normal distribution?
   a. $P(0 \le Z \le 1.50) = .4332$
   b. $P(Z \ge 2.0) = .0228$
   c. **$P(Z \ge -2.5) = .4938$ Page 230**
   d. $P(Z \le -1.0) = .1587$

174. The distribution function (df) is also known as cumulative distribution function (cdf).
   a. **Yes Page 173**
   b. No

175. Which of the following pairs of events are mutually exclusive?
   a. **A:the numbers above100;B: the numbers less than-200**
   b. A:the odd numbers; B:the number 5
   c. A:the even numbers; B:the numbers greater than 10
   d. A:the numbers less than 5; B:all the negative numbers

176. Two events A & B are said to be independent if...
   a. $P(A) + P(B)$
   b. $P(B/A) = P(B)$
   c. **$P(A) * P(B)$ Page 162**
   d. $P(A/B) = P(A)$

177. The collection of all outcomes for an experiment is called:
   a. **A sample space**
   b. Joint probability simple event
   c. The intersection of events
   d. Random experiment

178. Symbolically, a conditional probability is:
   a. P(AB)
   b. **P(A/B) Page 159**
   c. P(A)
   d. P(AUB)

179. Which of the following statements is INCORRECT about the sampling distribution of the sample mean?
   a. The standard error of the sample mean will decrease as the sample size increases

b. The standard error of the sample mean is measure of the variability of the sample mean among repeated samples
c. The sample mean is unbiased for the true (unknown) population mean
d. **The sampling distribution shows how the sample was distributed around the sample mean**

180. If one event is unaffected by the outcome of another event the two events are said to be:
a. Dependent
b. Not Mutually Exclusive
c. Mutually Exclusive
d. **Independent**

181. Let X be a random variable with binomial distribution, that is (X=0,1,…, n). The expected value E[X] is:
a. p
b. **np Page 214**
c. np(1-p)
d. Xnp

182. The sample mean is an unbiased estimator for the population mean. This means:
a. The sample mean has a normal distribution
b. **The average sample mean, over all possible samples, equals the population mean Page 259**
c. The sample mean is always very close to the population mean
d. The sample mean will only vary a little from the population mean

183. Probability of an impossible event is always:
a. Less than one
b. Greater than one
c. Between one and zero
d. **Zero Page 146**

184. The function abbreviated to d.f. is also called the......
a. Probability density function
b. Probability distribution function
c. **Commutative distribution function Page 173**
d. Discrete function

185. The total area under the normal curve is:
a. 0
b. **1 Page 186**
c. 0.5
d. 0.75

186. For exhaustive events, the P(AUBUC) is equal to:
a. P(A)
b. **P(S)** **Page 147**
c. P(A) * P(B)* P(C)
d. P(B)

187. One card is drawn from a standard 52 card deck. In describing the occurrence of two possible events, an Ace and a King, these two events are said to be:
Select correct option:
a. independent
b. randomly independent
c. random variables
d. mutually exclusive

188. The number of parameters in hypergeometric distribution is (are):]
a. 1
b. 2
c. **3** **Page 219**
d. 4

189. **Select correct option: If Y=bX, then variance of Y is**
a. **b*2 var(x)**
b. **var(x)**
c. **b var(x)**
d. **b square root var(x)**

190. **If f(x) is a continuous probability function, then P(X = 2) is:**
a. **1**
b. **0** **Page 186**
c. **1/2**
d. **2**

191. **Select correct option: In regression line Y=a+b X, Y is called:**
a. **Dependent variable** **Page 121**
b. **Independent variable**
c. **Explanatory variable**
d. **Regressor**

192. **If A and B are mutually exclusive events with P (A) =0.25 and P (B) = 0.50, Then P (A or B) =………**
a. **0.25**
b. **0.75** **Page 154**
c. **0.50**

d. 1

193. **In a 52 well shuffled pack of 52 playing cards, the probability of drawing any one diamond card is**

a. 1/52
b. 4/52
c. 13/52
d. 52/52

194. **Probability of a sure event is**

a. 8
b. 1 Page 154
c. 0
d. 0.5

195. **If Y=3X+5,then S.D of Y is equal to**

a. 9 s.d(x)
b. 3 s.d(x)
c. s.d(x)+5
d. 3s.d(x)+5

196. **The probability of drawing a red queen card from well-shuffled pack of 52 playing cards is**

a. 4/52
b. 2/52
c. 13/52
d. 26/52

197. **If P (B|A) = 0.25 and P (A and B) =0.20, then P (A) is**

a. 0.05
b. 0.80 Page 159
c. 0.95
d. 0.75

198. **A student solved 25 questions from first 50 questions of a book to be solved. The probability that he will solve the remaining all questions is:**

a. 0.25
b. 0.5
c. 1
d. 0

199. The classical definition of probability assumes:

a. Exhaustive events
b. Mutually exclusive events
c. **Equally likely evens Page 151**
d. Independent evens

200. In scatter diagram, the variable plotted along Y-axis is:
a. Independent variable
b. **Dependent variable**
c. Continuous variable
d. Discrete variable

201. Which of the following measures of dispersion are based on deviations from the mean?
a. Variance
b. Standard deviation
c. Mean deviation
d. **All of the these**

202. What does it mean when a data set has a standard deviation equal to zero?
a. All values of the data appear with the same frequency.
b. The mean of the data is also zero.
c. **All of the data have the same value.**
d. There are no data to begin with.

203. A set of possible values that a random variable can assume and their associated probabilities of occurrence are referred to as ________.
a. **Probability distribution**
b. The expected return
c. The standard deviation
d. Coefficient of variation

204. Which of the following can never be probability of an event?
a. 0
b. 1
c. 0.5
d. **-0.5**

205. The standard deviation of -1, -1, -1, -1 will be
a. 1
b. -1
c. **0**
d. Does not exist

206. The Special Rule of Addition is used to combine:
a. Independent Events
b. **Mutually Exclusive Events**
c. Events that total more than 1.00
d. Events based on subjective probabilities

207. set which is the sub-set of every set is
a. **Empty Set Page 134**
b. Power Set
c. Universal Set
d. Super Set

208. When two dice are rolled the number of possible sample points is :
a. 6
b. 12
c. 24
d. **36 Page 145**

209. If two events A and B are not mutually exclusive then
a. **P (A or B) = P (A) + P (B) – P (A and B) Page 157**
b. P (A or B) = P (A) + P (B)
c. P (A or B) = P (A) x P (B)
d. P (A or B) = P (A) + P (B)

210. Evaluate (10-4)!
a. 1000
b. **720**
c. 480
d. 32

211. When we toss a coin , we get only:
a. **1 outcome**
b. 2 outcome
c. 3 outcome
d. 4 outcome

212. If we roll a die then probability of getting a '6' will be
a. 2/6
b. **1/6**
c. 4/6
d. 1

213. If P(A) = 0.45, P(B) = 0.35, and P(A and B) = 0.25, then P(A | B) is:
a. 1.4
b. 1.8
c. **0.714 0.25/0.35 = 0.714 Page 159**

d. 0.556

214. Which of the following is not a measure of central tendency?

a. Percentile
b. Quartile
c. Standard deviation
d. Mode

215. Random experiment can be repeated any no. of times under the……… conditions.

a. Different
b. Similar Page 144

216. The simultaneous occurrence of two events is called:

a. Joint probability Page 194
b. Subjective probability
c. Prior probability
d. Conditional probability

217. In regression analysis, the variable that is being predicted is the

a. Dependent variable
b. Independent variable
c. Intervening variable
d. None of these

218. The probability of continuous random variable x on any particular point is always zero………………

a. Yes Page 186
b. No

219. P(an event) =no of favorable outcome/total no. of outcomes is the definition of

a. Subjective approach Page 149
b. Objective approach

220. If C is a constant ,then E(c)=

a. 0
b. 1
c. C Page 180
d. –c

221. When we toss a fair coin 4 times, the sample space consists of….points.

a. 4
b. 8

c. 12
d. 16

222. If we roll three fair dices then the total number of outcomes is:
a. 6
b. 36
c. 216 6*6*6=216
d. 1296

223. The probability of an event is always:
a. greater than 0
b. less than 1
c. between o and 1
d. greater than 1

224. In a multiplication theorem P (A and B) equals:
a. P(A)
b. P (B) P(A) + P (B)
c. P(A) * P (B|A) Page 159
d. P(B\A)*P(B)

225. If a die is rolled, what is the probability of getting an even number greater than 2?
a. 1/2
b. 1/3 2/6 = 1/3
c. 2/3
d. 5/6

226. In a Discrete probability distribution, P (x > 23) is read as:
a. P (there are more than 23 successes)
b. P (there are less than 23 successes)
c. P (there are at least 23 successes)
d. P (there are at most 23 successes)

227. A dormitory on campus houses 200 students. 120 are male, 50 are upper division students, and 40 are upper division male students. A student is selected at random. The probability of selecting a lower division student, given the student is a female, is:
a. 7/8
b. 7/20
c. 7/15
d. ¼

228. The function F(x) gives the probability of the event that X takes a value......
a. Less than x

b. Greater or equal to x
c. **Less or equal x Page 173**
d. Equal to x

229. In a simple regression line model, it is assume that the intercept parameter is equal to zero,

a. **The regression line will pass through the origin.**
b. The regression line will pass through the point (0,10)
c. The regression line will pass through the point (0,-10)
d. The slope of the line will also be zero.

230. If p(A n B)=p(A/B).p(B),then A and B are

a. Independent
b. Dependent
c. Equally likely
d. **Mutually exclusively**

231. A fair coin is tossed three times, the probability that at least one head appears,

a. 1/8
b. **7/8 (HHH,HHT,HTH, HTT, THH THT TTH TTT)**
c. 3/8
d. 5/8

232. A fair coin is tossed three times, the probability that at least one head appears,

a. 1/8
b. **7/8 (HHH,HHT,HTH, HTT, THH THT TTH TTT)**
c. 3/8
d. 5/8

233. In probability distribution, the sum of probabilities is equals to

a. 0
b. 0.1
c. 0.5
d. **1**

234. When a coin is tossed 3 times, the probability of getting 3 tails is

a. **1/8 (HHH,HHT,HTH, HTT, THH THT TTH TTT)**
b. 3/8
c. 3/6
d. 2/8

235. In how many ways can a team of 11 players be chosen from a total of 16 players?

a. **4368**

b. 2426
c. 5400
d. 2680

236. The standard deviation of c (constant) is
a. c
b. c square
c. 0
d. does not exist

237. Let E and F be events associated with the same experiment. Suppose the E and F are independent and that P(E) = 1/4 and P(F) = 1/2 Then P(E U F) is:
a. 1/8
b. **3/4** **Page 157**
c. 7/8
d. 5/8

238. In which of the following situations binomial distributions is approximate to normal distribution?
a. n = 50, p = 0.01
b. n = 500, p = 0.001
c. n = 100, p = 0.05
d. n = 50, p = 0.02

239. The location and shape of the normal curve is (are) determined by:
a. Mean
b. Variance
c. Mean & variance
d. **Mean & standard deviation**

240. A random experiment has five outcomes in its sample space {s1, s2, s3, s4, s5}. If P(s1)=0.2,P(s2)=0.3, P(s3)=0.1 and P(s4)=0.2 then P(s5)=?
a. 1
b. **0.2**
c. 0.8
d. 0.5

241. The joint density function f(x,y) will be a pdf if
a. **Both of integrals f(x,y) dx dy =1** **Page 200**
b. Both of integrals f(x,y) dx dy >1
c. Both of integrals f(x,y) dx dy <1
d. Both of integrals f(x,y) dx dy =0

242. Which of the following is correct property for joint probability distribution of X and Y:

a. **Sigma f(X,Y)=1 Page 194**
b. Sigma f(Y,X)=1
c. Both of above
d. None of above

243. A random variable that can assume every possible value in an interval [a, b]:
a. Discrete variable
b. **Continuous variable Page 9**
c. Qualitative variable
d. Categorical variable

244. Normal approximation to the binomial distribution is used when:
a. np>5
b. nq>5
c. **Both of the above Page 235**
d. None of the above

245. The Maximum Likelihood Estimators (MLE) are ………… and …………… but not necessarily ………
a. Unbiased, consistent, efficient
b. Consistent, unbiased, efficient
c. Unbiased, efficient, consistent
d. **Consistent, efficient, unbiased Page 264**

246. A probability density function 'f(x)' has the following property:
a. f(x) <= 0
b. f(x) < 0
c. **f(x) > 0 Page 186**
d. f(x) >= 0

247. For a continuous random variable X, P(X = x) is:
Select correct option:
a. **0 Page 186**
b. 0.5
c. 1
d. Undefined

248. An urn contains 4 red balls and 6 green balls. A sample of 4 balls is selected from the urn without replacement. It is the example of:
a. Binomial distribution
b. **Hypergrometric distribution Page 219**
c. Poisson distribution
d. Exponential distribution

249. The probability distribution of the proportion of successes in all possible samples is called the:
a. **Sampling distribution Page 237**
b. Sampling probability distribution
c. Sampling distribution of sample proportions
d. Sampling distribution of Population proportions

250. For good approximation of Poisson distribution to the binomial distribution, which of the following condition (s) is/are required:
a. The population size is large relative to the sample size
b. The probability, p, is close to .5 and the population size is large
c. **The probability, p, is small and the sample size is large Page 222**
d. The probability, p, is close to .5 and the sample size is large

251. If an estimator is more efficient then the other estimator, its shape of the sampling distribution will be
a. Flattered
b. Normal
c. **Highly peaked Page 261**
d. Skewed to right

252. Match the binomial probability P(x< 23) with the correct statement.
a. P(there are at most 23 successes)
b. **P (there are fewer than 23 successes)**
c. P (there are more than 23 successes)
d. P (there are at least 23 successes)

253. In statistics, the term 'expected value' implies the ___ value.
a. Independent
b. Normal
c. Standard
d. **Mean Page 191**

254. A quantity obtained by applying certain rule or formula is known as
a. **Estimate Page 264**
b. Estimator
c. Parameter
d. Proportion

255. Total no. of possible samples of size 2 (without replacement) from the population of size 6, will be:
a. 20
b. **15**
c. 10
d. 18

256. When a coin is tossed 3 times, the probability of getting 3 or less tails is

a. **1/2**
b. 0
c. 1
d. 3/5

257. Total no. of possible samples of size 3 (with replacement) from the population of size 6, will be:

a. 256
b. 196
c. **216 6^3**
d. 325

258. Using the normal approximation to the binomial distribution with n= 3 and p= 0.0571 the value of mean is:

a. **0.1713 Page 235**
b. 0.2132
c. 0.5133
d. 0.1923

259. Suppose 60% of a herd of cattle is infected with a particular disease. Let Y = the number of non-diseased cattle in a sample of size 5. the distribution of Y is:

a. Binomial with n = 5 and p = 0.6
b. **Binomial with n = 5 and p = 0.4**
c. Binomial with n = 5 and p = 0.5
d. Poisson with u = .6

260. The probability of success changes from trial to trial, is the property of:

a. **Binomial experiment Page 211**
b. Hypergeometric experiment
c. Both binomial & hypergeometric experiment
d. Poisson experiment

261. If a statistic used as an estimator, has its expected value equal to the true value of the population parameter being estimated then it is called........

a. Consistent
b. **Unbiased Page 258**
c. Efficient
d. Sufficient

262. In moments method, how many equations are needed to find the 2 unknown parameters?

a. **2** **Page 263**
b. 3
c. n/2

263. Which of the following is desirable for a good point estimator?
a. Consistency
b. Unbiasedness
c. Efficiency
d. **All of these** **Page 258**

264. The distribution function (df) is also known as
a. Probability distribution
b. Probability mass function
c. Probability density function
d. **Cumulative distribution function** **Page 173**

265. If X and Y are two discrete r.v's with joint probability function f(x,y), then the conditional probability function Y given X, f(y|x) is given by
a. **f(yj | xi) = f(xi, yj) / g(xi)** **Page 195**
b. f(yj | xi) = f(xi, yj) / h(yj)
c. f(yj | xi) = f(xi, yj) / Sum of g(xi)
d. f(yj | xi) = f(xi, yj) / sum of h(yj)

266. Which of the probability distributions has three parameters?
a. Binomial distribution
b. Normal distribution
c. **Hypergeometric distribution** **Page 219**
d. Poisson distribution

267. A standard deviation obtained from sampling distribution of sample statistics is known as
a. Sampling error
b. **Standard error** **Page 240**
c. Minimum error
d. Universal error

268. A parameter is a_____________ quantity.
a. Constant
b. **Variable**
c. Sample
d. Random

269. How can you define statistical inference?
a. **A decision, estimate, prediction or generalization about the population based on information contained in a sample**

b. A statement made about a sample based on the measurements in that sample
c. A set of data selected from a larger set of data
d. A decision, estimate, prediction or generalization about sample based on information contained in a population

270. Which of the following is most important and most widely used method in point estimation?
a. The method of moments
b. The method of fractional moments
c. **The method of leas square Page 263**
d. The method of maximum likelihood

271. Binomial distribution is skewed to the right if:
a. p=q
b. **P<q Page 215**
c. p>q
d. p=n

272. In a discrete distribution function, F(23) can be stated as:
a. P (there are at most 23 successes)
b. P (there are at least 23 successes)
c. P (there are less than 23 successes)
d. P (there are more than 23 successes)

273. The Probabilities of the various values of the sample satistic can be computed using the......... definition of probability.
a. Subjective
b. Objective
c. **Classical Page 149**
d. None of the baove

274. E(10X +3) =_______
a. 10 E(X)
b. E(X)+3
c. **10 E(X)+3**
d. 100E(X)

275. If p is very small and n is considerably large then we shall apply the:
a. Binomial distribution
b. Hypergeometric distribution
c. **Poisson distribution Page 222**
d. Exponential distribution

276. Which of the following is NOT applicable to a Poisson distribution?
a. **IF P = 0.5 & n = 19**

b. IF P = 0.01 & n = 200
c. IF P = 0.02 & n = 300
d. IF P = 0.03 & n = 500

277. The normal distribution has points of infection which are equidistance from the:
a. Median
b. Mean Page 229
c. Mode
d. Mean,Median & Mode

278. If a random variable X denotes the number of heads when we toss a fair coin 5 times, the X assumed the values:
a. 0,1,2,3
b. 1, 2,3,4,5
c. 0, 1, 2,3,4,5
d. 1, 5, 5

279. As a rule of thumb, when n>=30, then we can assume that………is normally distributed:
a. Probability distribution
b. Sampling distribution Page 243
c. Binomial distribution
d. Sampling distribution of sample mean

280. If b(x, 7, 0.30), the variance of this distribution is:
a. 1.77
b. 1.74
c. 1.44
d. 1.47 Page 333

281. Which of the following is a characteristic of a binomial probability experiment?
a. Each trial has more than two possible outcomes
b. P(success) = P(failure)
c. Probability of success changes for each trail
d. The result of one trial does not affect the probability of success on any other trial Page 211

282. The number of parameters in uniform distribution is (are):
a. 1
b. 2 Page 225
c. 3
d. 4

283. The probability can never be:

a. -1
b. 1/2
c. 1
d. **-1/2**

284. The conditional probability P (A\B) is:
a. **P(A n B)/P(B)** **Page 159**
b. P(A n B)/P(A)
c. P(A U B)/P(B)
d. P(A U B)/P(A)

285. The binomial distribution is negatively skewed when:
a. **p>q** **Page 215**
b. p<q
c. p=q
d. p=q=1/2

286. When we draw the sample with replacement, the probability distribution to be used is:
a. **Binomial**
b. Hypergeometric
c. Binomial & hypergeometric
d. Poisson

287. The moment ratios of normal distribution come out to be:
a. 0 and 1
b. 0 and 2
c. **0 and 3** **Page 227**
d. 0 and 4

288. Suppose the test scores of 600 students are normally distributed with a mean of 76 and standard deviation of 8. The number of students scoring between 70 and 82 is:
a. 272
b. 164
c. 260
d. **328**

289. If P (A) = 0.3 and P (B) = 0.5, find P (A/B) where ‘A’ and ‘B’ are independent:
a. 0.3
b. 0.5
c. 0.8
d. **0.15** **Page 162**

290. If the second moment ratio is less than 3 the distribution will be:

a. Mesokurtic
b. Leptokurtic
c. **Platykurtic**
d. None of these

291. For the independent events A and B if P (A) = 0.25, P (B) =0.40 then P (A and B)=……

Select correct option:
a. 0.65
b. **0.1** **Page 162**
c. 0.50
d. 0.15

292. A random variable X has a probability distribution as follows: X | 0 1 2 3 P(X) | 2k 3k 13k 2k What is the possible value of k:
a. 0.01
b. 0.03
c. **0.05**
d. 0.07

293. The probability of drawing any one spade card is:
a. 1/52
b. 4/52
c. **13/52**
d. 52/52

Question # 1

A standard deviation obtained from sampling distribution of sample statistics is known as

- Sampling error
- Standard error (Page 240)
- Minimum error
- Universal error

Question # 2

The probability distribution of the proportion of successes in all possible samples is called the:

- Sampling distribution (Page 237)
- Sampling probability distribution
- Sampling distribution of sample
- proportions Sampling distribution of

Question # 3

Conventionally, the probability of making a type-II error is denoted by:

- Beta Lect. 36, page no. 273 handouts
- Sigma
- Alpha
- Theta

Question # 4

Conventionally, the probability of making a type-I error is denoted by which of the following symbol?

- Theta
- Alpha Lect. 36, page no. 273
- Sigma
- Beta

Question # 5

Which of the following is most important and most widely used method in point estimation?

- The method of moments
- The method of fractional moments
- The method of maximum likelihood
- The method of least square

Question # 6

106. A random sample of n = 6 has the elements 6, 10,13,14,18 and 20.What is the point estimate of the population mean?

- 12
- 13.5
- 11
- 11.5

Question # 7

The test statistic to test the U1 = U2 (U represent the mean of population) for normal population for n>30.

- F-test
- Z-test
- T-test
- Chi-Square test

Question # 8

If a significance level of 1% is used rather than 5%, the null hypothesis is:

- More likely to be rejected
- Less likely to be rejected
- Just as likely to be rejected
- None of the above

Question # 9

To determine a sample size which test we use .

- Z
- T
- None of these
- F

Question # 10

We can apply method of Maximum Likelihood on:

- Discrete variables only
- Discrete as well as continuous variables
- Continuous variables only
- Qualitative variables only

Question # 11

"A point estimate plus/minus a few times the standard error of that estimate". This statement represents:

- Confidence interval
- Critical value
- Acceptance region
- Critical region

Question # 12

Conventionally, the probability of making a type-II error is denoted by:

- Beta
- Sigma
- Alpha
- Theta

Question # 13

A deserving player was not selected in the cricket team. It is an example of:

- Type-II error
- Type-I error
- Type-IV error
- Type-II error

Question # 14

"To consider every possible value that the parameter might have, and for each value, compute the probability and THAT value of the parameter for which the probability of a given sample is greatest, is chosen as an estimate." This procedure is known as

- The method of moments
- The method of least square
- The method of maximum likelihood
- The method of fractional moments

Question # 15

When sample size is to be considered large, population standard deviation can be replaced by:

- Sample standard deviation
- Population variance
- Sample mean
- Sample variance

Question # 16

An estimator is said to beif its expected value is equal to true value of its parameter.

- Efficient

- Biased
- Unbiased
- Consistent

Question # 17

Normal approximation to the binomial distribution is used when:

- np>5
- nq>5
- Both of the above (Page 235)
- None of the above

Question # 18

Interval estimation and Confidence interval are:

- Same
- Different
- Dependent
- Independent

Question # 18

An experiment was conducted to estimate the mean yield of a new variety of wheat. A sample of 20 plots gave a mean yield of 2.9, and a 95% confidence interval of (2.48, 3.32). This means:

- We are 95% confident that the true mean yield of this variety is between 2.48 and 3.32
- We are 95% confident that the true mean yield of this variety is 2.9
- None of these
- We are sure the true mean of this new variety is between 2.48 and 3.32

Question # 19

If Var (mean1) < Var (mean2) it implies

- Mean 1 is more consistent
- Mean 2 is more consistent
- Mean 2 is more efficient
- Mean 1 is more efficient

Question # 20

When sample size is to be considered large, population standard deviation can be replaced by:

- Sample standard deviation
- Population variance
- Sample mean
- Sample variance

Question # 21

If an estimator is more efficient than the other estimator, its shape of the sampling distribution will be

- Flattered
- Skewed of right
- Normal
- Highly peaked

Question # 22

In interval estimation we obtained a______of values as an estimate of parameter.

- Group
- Range
- Both a and b
- None of these

Question # 23

Alpha is the probability of:

- Type-I error
- Committing type-I error
- All of these
- Reject H0/H0 is true

Question # 24

Which of the following is not criterion for a good problem situation?

- It must be clearly structured and defined
- It must be appropriate to the students' level of development

- It must be authentic
- It must be such that solution will be benefited by group effort

Question # 25

Total no. of possible samples of size 2 (without replacement) from the population of size 6, will be:

- 20
- 15
- 10
- 18

Question # 24

In binomial distribution the sample size is considered to the sufficiently large, if both np and nq are greater than or equal to:

- 10
- 3
- 5
- 15

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